Algebra 2 — Algebraic fractions · Roots and powers · Indices · Inequalities

No calculator. Marks in brackets. ★ = harder question. Check every solution by substitution.

A · SIMPLIFY AS MUCH AS POSSIBLE

1 $\dfrac{15x^2y}{5x}$ [1]

2 $(8ab^2) \div (12a^2b)$ [1]

3 $\dfrac{a}{b^2} \times \dfrac{b^3}{c} \times \dfrac{c^2}{a}$ [1]

4 $\dfrac{4x^2}{y} \div \dfrac{x}{y^2}$ [1]

5 $\dfrac{x}{3} - \dfrac{x}{5}$ [2]

6 $\dfrac{x}{4} + \dfrac{3x}{8}$ [2]

B · EQUATIONS WITH POWERS AND ROOTS

7 Solve $3x^2 - 5 = 70$ [3]

8 Solve $\dfrac{x^2 - 6}{2} = 15$ [3]

9 Solve $\dfrac{\sqrt{x - 3}}{2} = 2$ [2]

C · INDICES

10 Simplify [3]

a $2(k^4)^3$ **b** $m^7 \div m^2$ **c** $t^3 \times t^9$

D · INEQUALITIES

11 Solve the inequality $12 \leq 9 - x$ and show the solution on a number line. [3]

12 List the integer solutions of $2 \leq 2x < x + 7$. [3]

13 Solve $3(x - 2) > 5(x - 4)$ [2]

14 Solve $-2x + 5 \geq 11$ [2]

Total: 29 marks

Check: cancel only common *factors* · $x^2 = 25$ has two solutions · multiplying or dividing an inequality by a negative flips the sign · solid circle for $\leq$, open circle for $<$.